ModelIt!

Teacher's Guide

Why Do Objects Move Differently?

Balanced and Unbalanced Forces

Quick Reference

Grade Level	Grade 3
NGSS Standard	3-PS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 3-PS2-1

Plan and conduct an investigation to provide evidence of the effects of balanced and unbalanced forces on the motion of an object.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students plan and conduct an investigation to provide evidence of how balanced and unbalanced forces affect the motion of an object across different surfaces.
Disciplinary Core Idea	**PS2.A Forces and Motion** — Each force acts on one particular object and has both strength and a direction. An object at rest typically has multiple forces acting on it, but they add to give zero net force on the object. If the total force on the object is not zero, its motion will change.
Crosscutting Concept	**Cause and Effect** — Students investigate how changing the push force (cause) and surface friction (cause) affect the motion of objects (effect) in a predictable pattern.

CAST Testing Alignment

MULTIPLE CHOICE:

A student pushes a block across a smooth table and then across a rough carpet with the same force. The block slides farther on the table. Which best explains why?

CONSTRUCTED RESPONSE:

A toy car is sitting still on a flat table. You give it a push and it rolls across the table and eventually stops. Using what you know about forces, explain why the car started moving and why it stopped. Use the words 'unbalanced forces' and 'friction' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Forces are everywhere — every push, pull, twist, and turn involves forces. Understanding how forces interact helps us explain why things move the way they do, from a rolling ball to a flying airplane.

What Is a Force?

A force is any push or pull on an object. Forces have two important properties: strength (how hard the push or pull is) and direction (which way the push or pull goes). Forces can make objects start moving, stop moving, speed up, slow down, or change direction. Gravity is a force that pulls everything toward Earth. Friction is a force that resists motion when surfaces rub together. Magnetic force pulls certain metals. Forces are measured in units called Newtons.

Balanced vs. Unbalanced Forces

When all the forces on an object add up to zero, the forces are balanced and the object does not change its motion. A book on a table has balanced forces — gravity pulls down and the table pushes up equally. When forces do NOT add up to zero, they are unbalanced, and the object's motion changes. Kicking a ball applies an unbalanced force — the kick is much stronger than friction, so the ball moves. Two people pushing a box with equal force from opposite sides create balanced forces — the box stays still.

Friction: The Hidden Force

Friction is a force that opposes motion whenever two surfaces touch and slide against each other. Rough surfaces create more friction than smooth ones — that is why it is harder to slide on carpet than on ice. Friction converts kinetic energy into heat energy, which is why rubbing your hands together makes them warm. While friction can be annoying (it slows things down), it is also essential — without friction, you could not walk (your feet would slip), drive (tires would spin), or even hold a pencil (it would slide out of your hand).

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify push force as an external component and friction and object motion as internal components that interact to determine how the object moves.

E **Establish Relationships**
Students determine that push force positively affects object motion (bigger push means more movement) while friction negatively affects it (more friction means less movement). Object motion depends on the balance between push and friction.

V **Visualize & Model**
Students build a model showing how push force and friction interact to determine the speed and distance of object motion.

E **Evaluate Outcomes**
Students run scenarios with different push forces on different surfaces to observe how changing both variables affects object motion across the system.

R **Revise & Extend**
Students add object weight or surface type to explore how heavier objects and different materials change the balance of forces and resulting motion.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Push your pencil across your desk right now. Why did it move? Now push down on your desk — why did it NOT move?"*

Watch For

Have students push various objects and observe what happens. Collect observations: What makes things move? What keeps things still?

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to figure out the secret rules that control why things move — or stay perfectly still!"*

Watch For

Read objectives together. Introduce vocabulary with demos: push a book (force), rub hands (friction), tug-of-war (balanced/unbalanced).

Slide 3: The Big Question

3 min

Say: *"Why does a ball roll when you push it but a book on a table just sits there? Same world, same forces — so what is different?"*

Watch For

Turn and talk: If everything has forces on it, why do some things move and some do not? Collect theories.

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to test how pushes and friction work together to control motion!"*

Watch For

Preview each LEVER step. Do a quick tug-of-war to demonstrate balanced (tied) and unbalanced (one side wins) forces.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What parts of this system can we identify? What forces are at work when we push something?"*

Watch For

Guide sorting of three components. Explain push force is external because WE decide how hard to push, while friction and motion happen inside the system.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When you push harder, does the object move more or less? And what does friction do to motion?"*

Watch For

Help students draw a positive arrow from push force to object motion, and discuss how friction opposes motion.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Experiment time! Same push, different surfaces — and then same surface, different pushes. What will happen?"*

Watch For

Have students predict first, then run each scenario. Compare results: smooth vs. rough, gentle vs. hard push.

Slide 8: What Did We Discover?

7 min

Say: *"Now we know the secret rules of motion — it is ALL about whether forces are balanced or unbalanced!"*

Watch For

Summarize discoveries. Final demo: push a book on the table (unbalanced force makes it move), then let go (friction balances the push and it stops).

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Push Force
- **INTERNAL:** Friction, Object Motion

Relationships

- **Push Force to Object Motion:** POSITIVE (+) — A stronger push applies more force to the object, making it move faster and farther. Double the push force and the object covers significantly more distance. The push gives the object energy to move.
- **Friction to Object Motion:** POSITIVE (+) — Note: friction OPPOSES motion, but in the model, as friction increases, its EFFECT on slowing motion increases. More friction means the object slows down faster and stops sooner. Rough surfaces create high friction that fights against the push force.

Key Discoveries

- Objects only move when forces are UNBALANCED — the push must be stronger than friction for the object to move
- Stronger pushes create more motion because the force is more unbalanced — the push overcomes friction easily
- Rough surfaces create more friction, which slows objects down and makes them stop sooner
- A book on a table stays still because the forces are balanced — gravity pulls it down and the table pushes it up equally

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Objects need a constant push to keep moving"	Once an object is moving, it will keep moving until another force (like friction or a wall) stops it. This is Newton's first law of motion — an object in motion stays in motion unless acted on by an outside force. A hockey puck slides across ice for a long time because there is very little friction to stop it. You do not need to keep pushing!
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heavier objects always move slower"	Weight does not determine speed — force and friction do. A heavy bowling ball and a light tennis ball dropped from the same height fall at the same speed (ignoring air resistance). On a flat surface, a heavier object needs more force to START moving, but once moving, it can travel just as fast or faster than a lighter one if friction is the same.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Friction is always bad and we should try to eliminate it"	Friction can slow things down, which sometimes seems annoying. But friction is ESSENTIAL for everyday life. Without friction, you could not walk (your feet would slip), cars could not drive (tires would spin), and you could not hold anything (everything would slide out of your hands). Brakes use friction to stop cars. Shoes use friction to grip the ground. Friction is a HELPER, not just a nuisance.